Section 1
Foundations of Quantum Machine Learning

QWorld QML Tutorial

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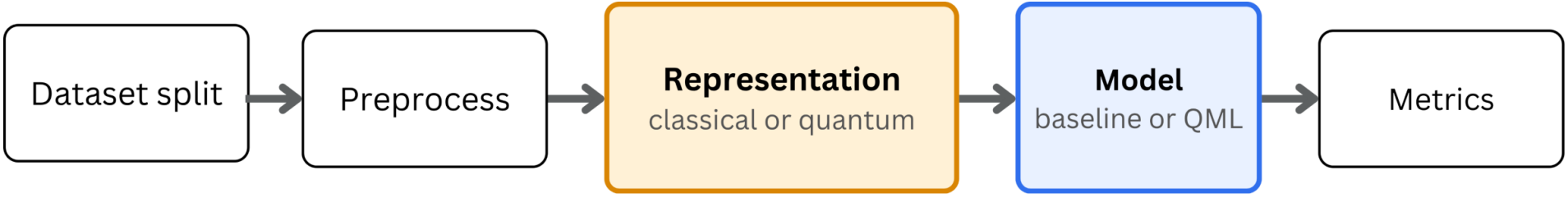
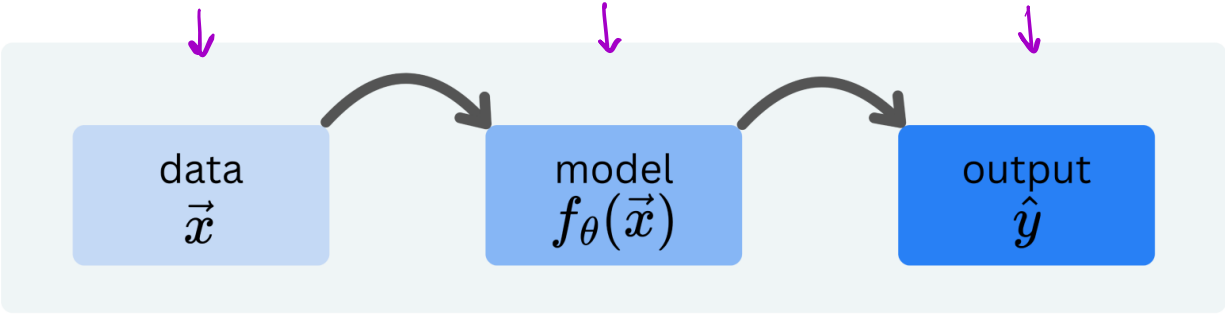
02 What This Lecture Covers

- what QML means at a practical level,
- how to think about learning tasks and evaluation,
- why strong classical baselines come before quantum claims.
- deeper ansatz design comes later,
- training algorithms in detail come later,
- quantum advantage claims are not the goal of this section.



03 What Is Machine Learning?

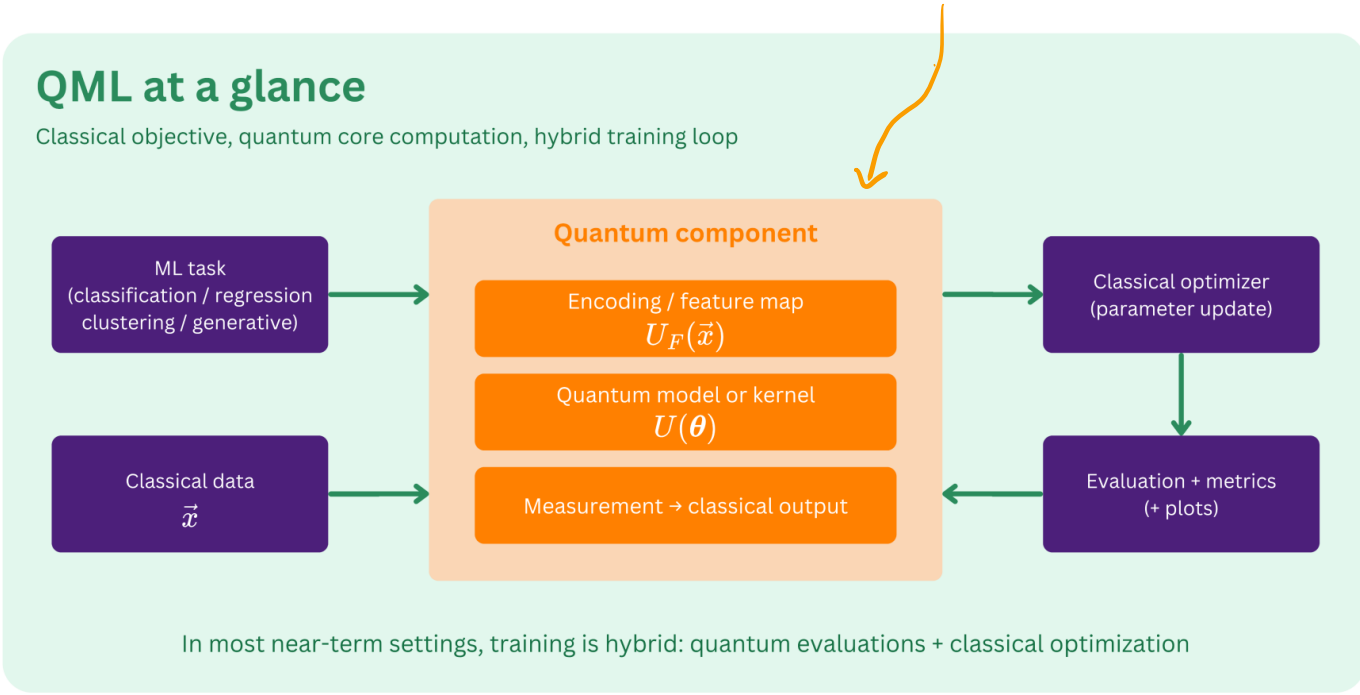
- ML learns useful patterns from data to make predictions, decisions, or generate samples.
- The important order is:
  - define the task,
  - identify the input and desired output,
  - then choose a model family that can support that task.
- This matters for the rest of the tutorial because QML should be understood as one possible modeling approach inside ML, not as a separate replacement for task design.
- The slide should leave the audience with the simple abstraction:
  - data in,
  - model with parameters,
  - prediction out.



$$\hat{y} = f_\theta(\vec{x})$$

04 What Is Quantum Machine Learning?

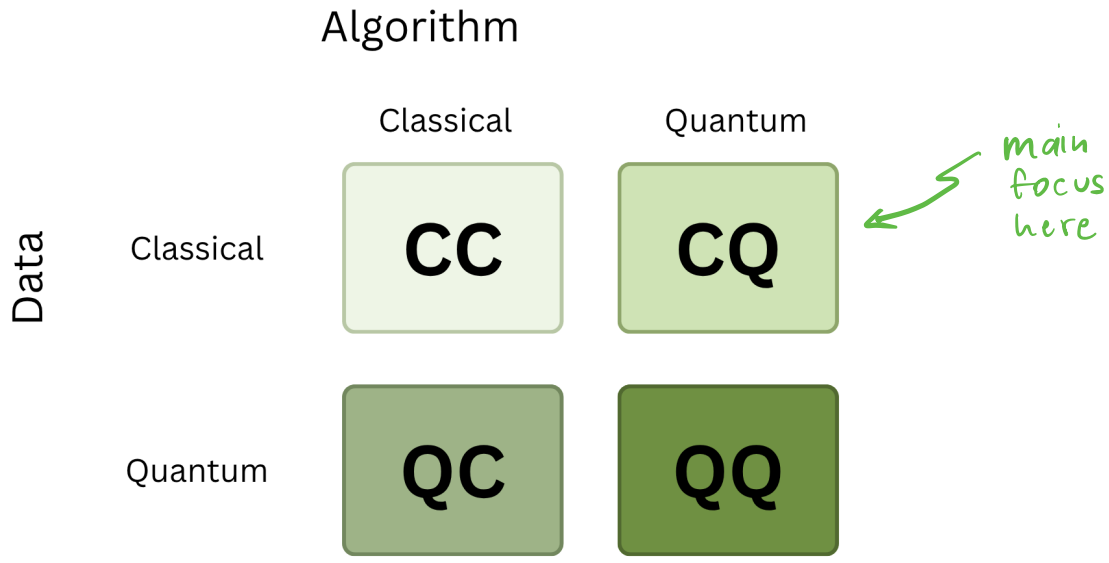
- QML keeps the learning objective classical but inserts quantum components into the pipeline.
- In practice, near-term QML is usually hybrid quantum-classical.
- The output we care about is still
- What changes is the internal representation or model component:
  - data may be encoded into a quantum state,
  - a quantum circuit or kernel may process that representation,
  - and a measurement converts the result back into classical information.



$$\vec{x} \xrightarrow{U_F(\vec{x})} |\psi(\vec{x})\rangle \xrightarrow{\text{quantum model } U(\boldsymbol{\theta})} \text{measurement} \rightarrow \hat{y}$$

05 Four Data–Algorithm Regimes

- QML is not only one setting.
- It is useful to distinguish whether the data are classical or quantum, and whether the algorithmic core is classical or quantum.
- This tutorial focuses mainly on the CQ regime: classical data processed with quantum representations or models.



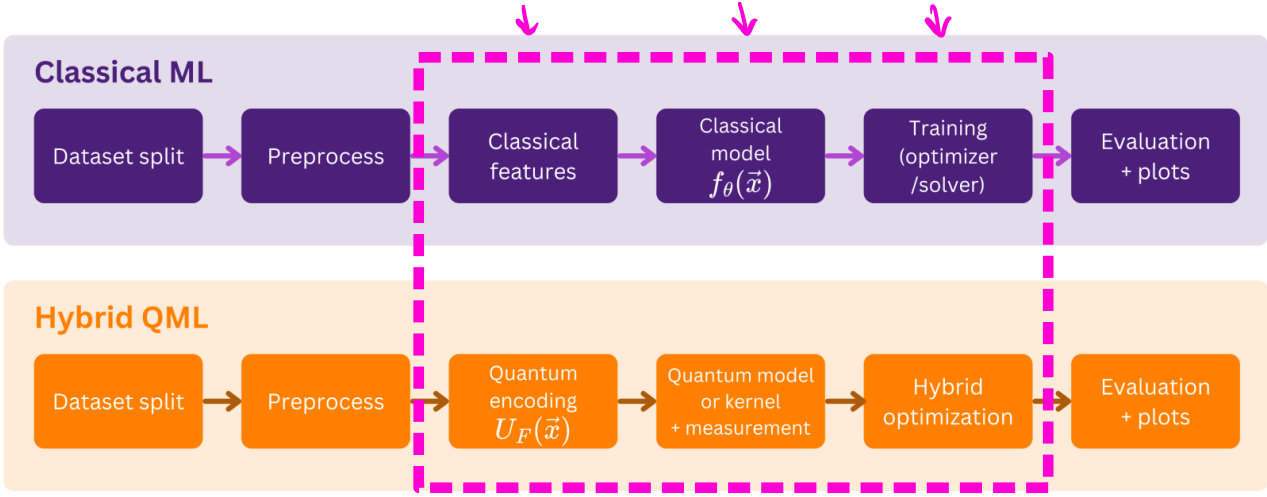
06 Hybrid Does Not Mean Automatic Advantage

- A hybrid pipeline is not automatically better just because it contains a quantum circuit.
- Serious QML requires a fair baseline, a fair protocol, and a concrete hypothesis.

Needed before any serious performance claim or discussion of possible advantage

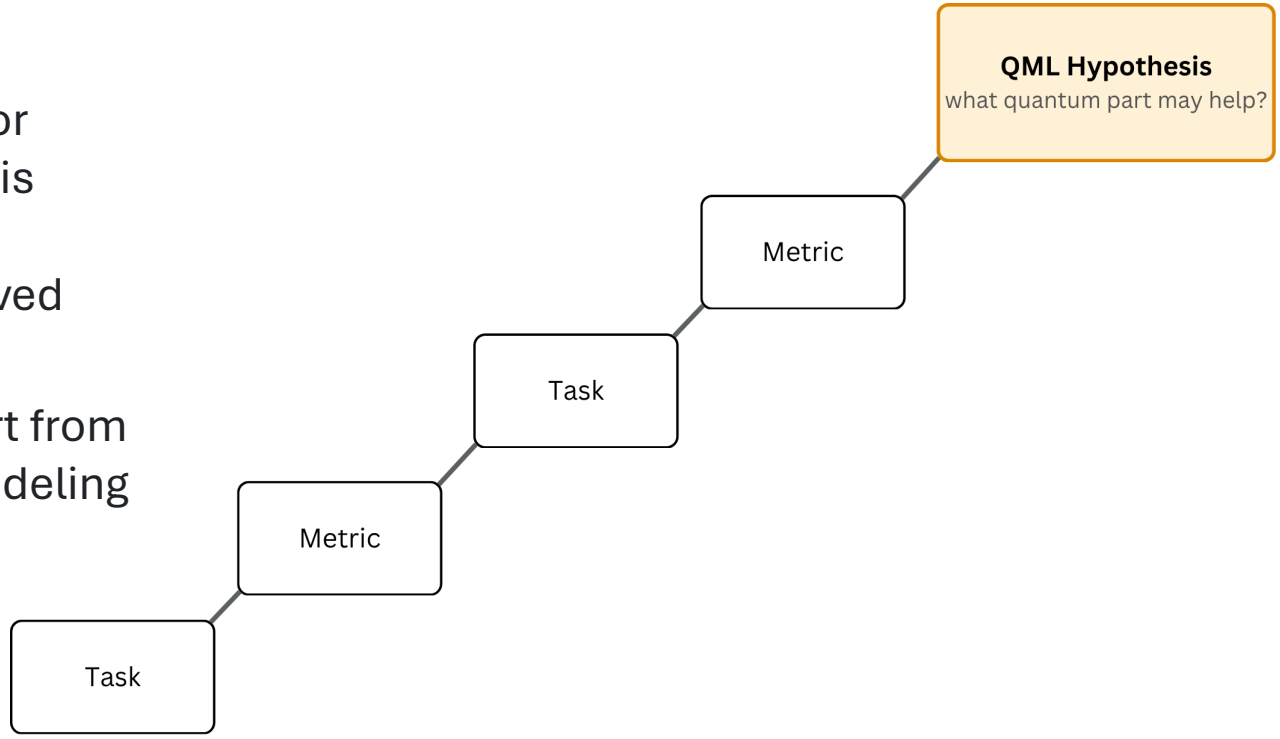
07 Classical Pipeline vs Hybrid QML Pipeline

- Most of the ML pipeline stays the same.
- In hybrid QML, the representation/model/output stage is where the quantum part often enters.
- This is one of the most important organizational ideas of Section 1.
- Students should see that dataset splitting, preprocessing, evaluation, and comparison discipline do not disappear in QML.
- The real change is more local:
- how data is represented,
- how the model produces an output,
- and what quantum object is measured.
- That is why a strong ML foundation is not optional background; it is part of doing QML correctly.



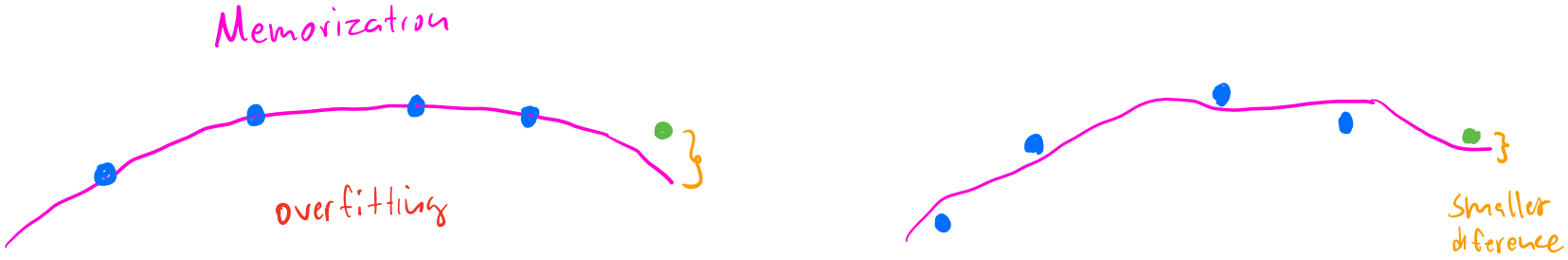
08 Why Start with Classical ML in a QML Course?

- Because QML only makes sense when the task, metric, and comparison protocol are already well defined.
- A baseline is a reference point for judging whether a QML approach is actually adding value; it is not a statement that QML must be derived from a classical model.
- The notebooks intentionally start from ML discipline before quantum modeling so that later QML claims are scientifically interpretable.



09 Generalization, Not Memorization

- In both ML and QML, the real objective is generalization: performing well on unseen data, not memorizing the training set.
- A model that only memorizes is not useful scientifically or practically, even if its training performance looks excellent.
- This is one of the methodological foundations of all later ML and QML work in the tutorial.



training performance ≠ generalization performance

10 When QML Might Be Worth Considering

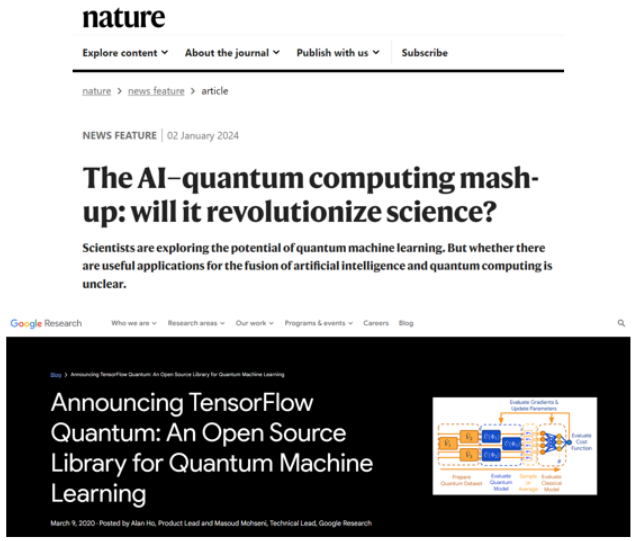
- QML is worth exploring only when several positive conditions line up at the same time.
- This is an entry checklist, not a performance claim.
- A good reason to consider QML is not “because quantum is interesting”.
- A good reason is that the task, data geometry, or modeling hypothesis gives a concrete motivation for trying a quantum representation or quantum model.
- The slide should sound disciplined rather than promotional:
- QML is a candidate when there is a meaningful technical question to test,
- not when one is simply hoping for improvement without a hypothesis.

Explore QML when you can answer “yes” to questions like these:

1. The task or data has structure that a richer representation might exploit.
2. There is a concrete quantum hypothesisabout encoding, kernel geometry, or model behavior.
3. The problem scale is feasiblefor the encoding, simulator, or hardware path you plan to use.
4. A solid reference protocol existsso that any future QML claim can be interpreted fairly.

Why Study QML Now?

- A field can be scientifically worth studying even before its practical usefulness is fully established.
- QML is still early, but early-stage research can shape future methods, tools, and ways of thinking.



- Nature News Feature (2024), The AI–quantum computing mash-up: will it revolutionize science?: https://www.nature.com/articles/d41586-023-04007-0
- CERN News (2023), `Shaping future quantum techniques in machine learning at CERN: https://home.cern/news/news/computing/shaping-future-quantum-techniques-machine-learning-cern
- Google Research Blog (2020), Announcing TensorFlow Quantum: An Open Source Library for Quantum Machine Learning: https://research.google/blog/announcing-tensorflow-quantum-an-open-source-library-for-quantum-machine-learning/
- Alpha Events / Quantum Tech (industry article), Rahko: The future of pharmaceuticals is Quantum machine learning: https://www.alphaevents.com/events-quantumtechus/blog/rahko-the-future-of-pharmaceuticals-is-quantum-machine-learning
- Nature Photonics (2025), Experimental quantum-enhanced kernel-based machine learning on a photonic processor: https://www.nature.com/articles/s41566-025-01682-5
- npj Quantum Information (2025), Entanglement-induced provable and robust quantum learning advantages: https://www.nature.com/articles/s41534-025-01078-x
- Nature Communications (2025), Quantum advantage for learning shallow neural networks with natural data distributions: https://www.nature.com/articles/s41467-025-68097-2
- npj Quantum Information (2026), Exponential quantum advantages in learning quantum observables from classical data: https://www.nature.com/articles/s41534-025-01162-2

12 When QML Is Probably Not the Right First Tool

- Some setups are not ready for QML yet, even if the topic sounds attractive.
- This slide is about warning signs and stop conditions, not about balanced pros and cons.
- The point is not to dismiss QML as a field.
- The point is to say that some problem setups are not mature enough for QML to be a responsible first move.
- If there is no quantum hypothesis, no credible evaluation path, or no realistic encoding story, the better next step is to improve the problem formulation first.
- This slide should feel like methodological caution:
- do not start with the most exotic tool if the experimental logic is still weak.

1. No clear quantum hypothesis
2. Baseline dominates the problem
3. Encoding unrealistic
4. Evaluation discipline too weak

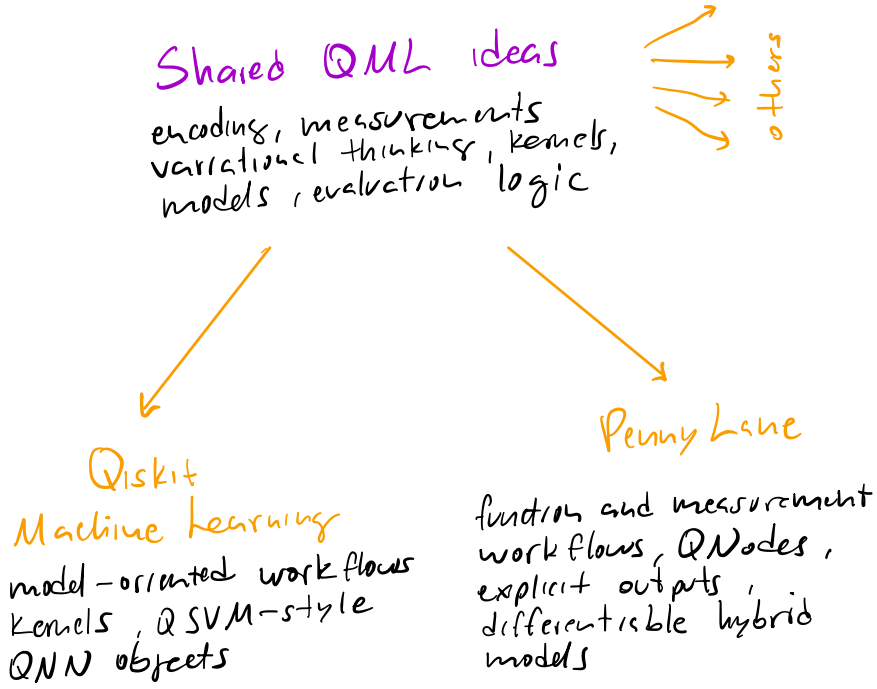
↓

Better next step

- Tighten the task
- Strengthen the baseline
- Revisit QML later

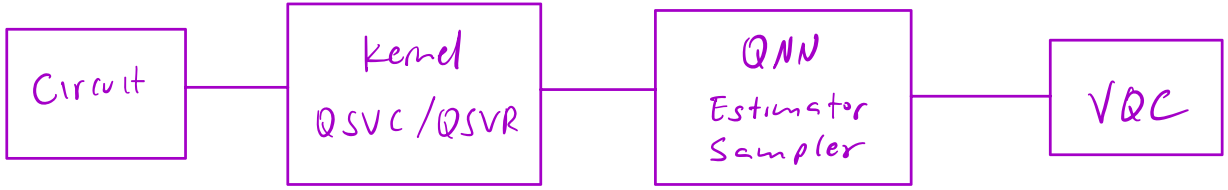
13 Two Frameworks, Two Styles

- The tutorial uses both Qiskit Machine Learning and PennyLane, not only this section.
- One reason is technical literacy: students should see that the same QML ideas can live inside different software abstractions.
- Another reason is practical choice: some workflows feel more natural in a model-oriented ecosystem, while others feel more natural in a function-and-measurement ecosystem.
- A third reason is pedagogical: learning both helps separate the core ideas from the API details of any single framework.



Qiskit Machine Learning: Model-Oriented View

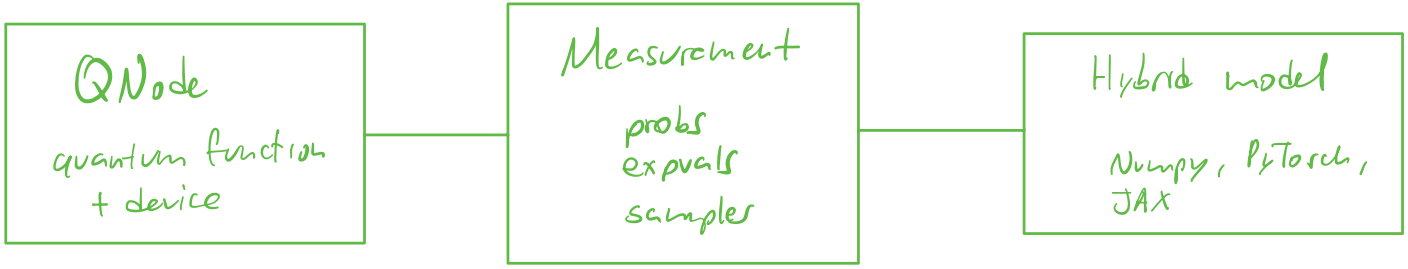
- Qiskit ML feels close to choosing a model family and then assembling the needed quantum ingredients.
- Kernels, QSVM-like workflows, QNN objects, and estimator-style interfaces are central.



Kernel: $k(\bar{x}, \bar{x}')$

PennyLane: Function-and-Measurement View

- PennyLane starts naturally from a quantum function and an explicit measurement choice.
- The QNode is the organizing abstraction.



$$f_\theta(\bar{x}) = \text{QNode}(\bar{x}, \theta) \rightarrow \text{measurement}$$

16 Why Measurements Matter So Much

- The measurement determines what classical information comes out of the circuit.
- That output choice shapes what task the model can support, how we interpret it, and what loss or metric makes sense.
- So measurement is part of model design, not only the final readout step.

Same quantum circuit

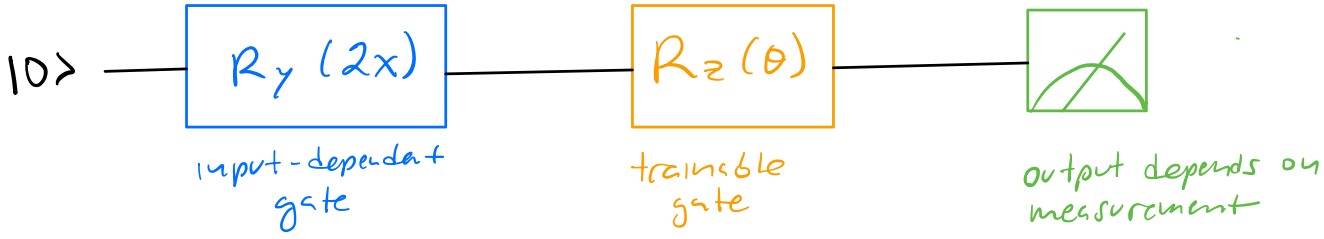
probabilities
$p(0)$, $p(1)$
become class scores
useful for classification

expectation values
$\langle Z \rangle$, $\langle X \rangle$
give scalar outputs
useful as regressors

samples
repeated draws
preserve stochastic structure
useful for generative settings

17 Same Variational Idea, Two Frameworks

- The point of the small one-qubit example is not expressive power.
- The point is that both frameworks implement the same quantum circuit, but expose it through different software abstractions.



$$|\psi(x,\theta)\rangle = R_z(\theta)\, R_y(2x)\, |0\rangle$$

18 Input Encoding vs Trainable Parameters

- **x** is the classical feature,
- **θ** is the trainable parameter,
- but those two roles are not interchangeable.
- The input-dependent part tells the circuit what sample it is processing.
- The trainable part tells the model how it is allowed to respond across many samples.
- In the toy one-qubit example, that distinction is visible because **x** changes the measured probability directly, while **θ** affects a phase that this specific measurement does not turn into a probability change.

Data path

sample-dependent information enters here

$R_y(2x)$

encoding gate

changes from sample to sample

Trainable path

model-dependent flexibility enters here

$R_z(\theta)$

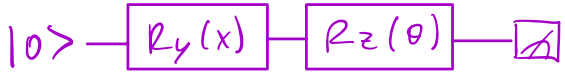
trainable gate

shared across many samples

19 A Differentiable Quantum Circuit

- A circuit is differentiable when its measured output changes in a controlled way as trainable parameters change.
- That matters because hybrid learning often relies on adjusting parameters step by step, not by guessing them from scratch.
- The goal here is not to teach gradient methods yet, only to explain why differentiability becomes valuable once training enters the picture.
- Later sections will return to this idea in a more concrete way when optimization and variational training are introduced.

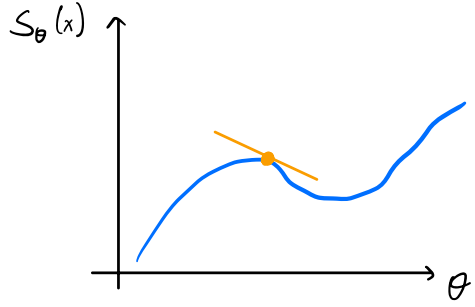
Parameterized circuit



circuit output : $s_\theta(x)$ = measured quantity

(ex.: probability or expval)

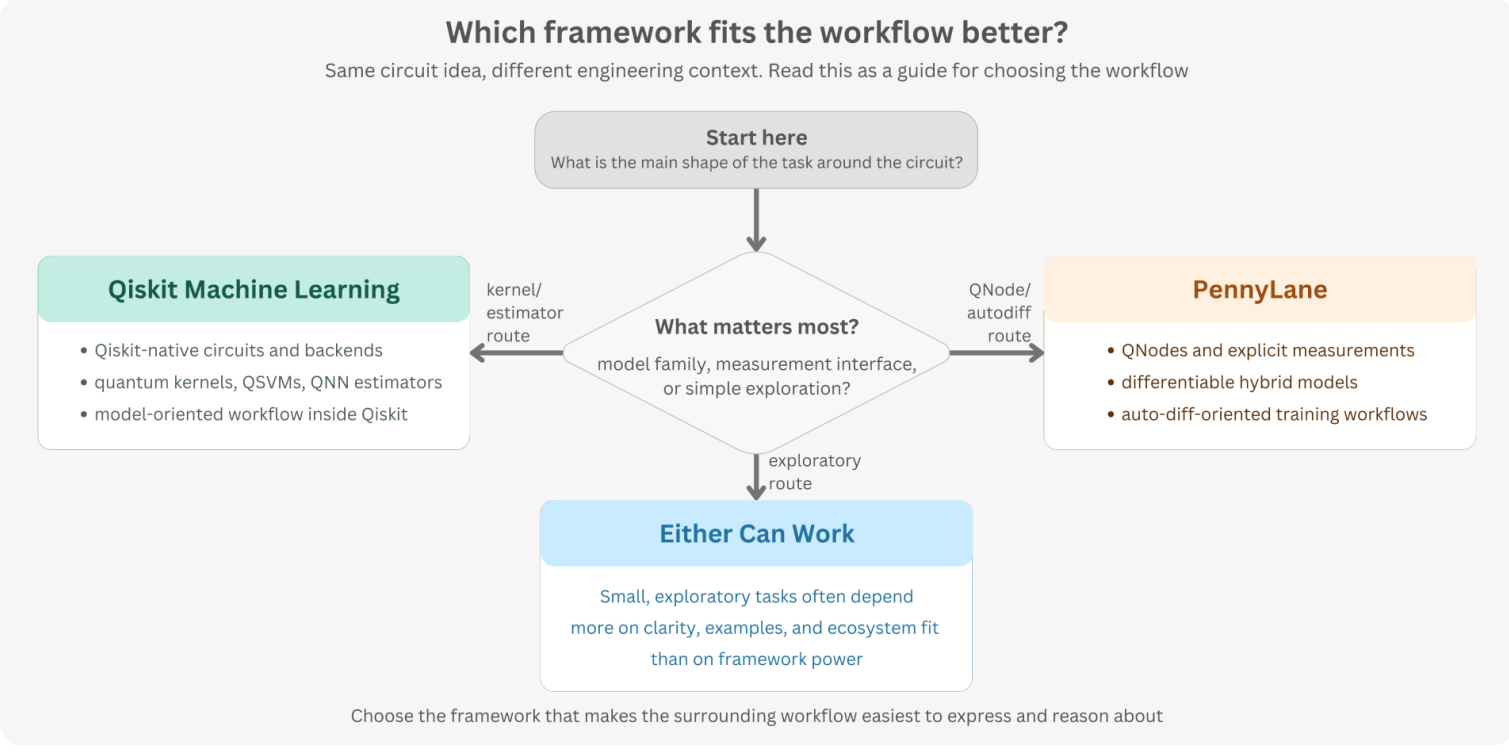
Measured output as a function of θ



$$\frac{\partial s_\theta(x)}{\partial \theta}$$

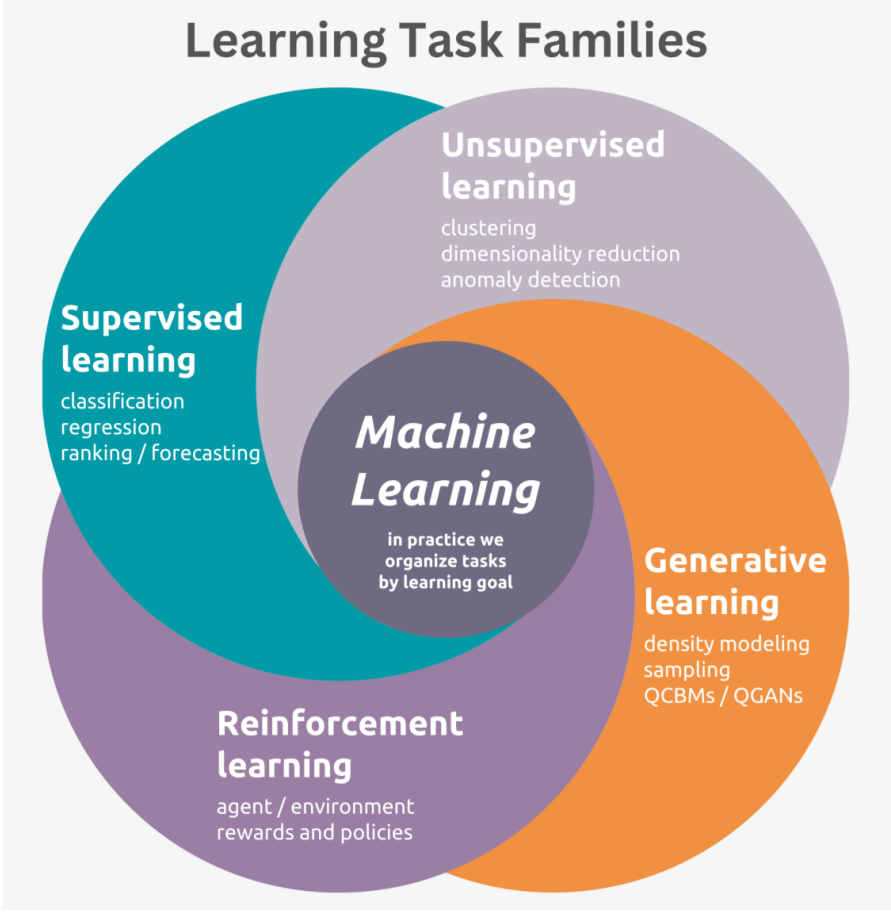
20 Framework Selection Is Task-Dependent

- There is no universal winner between the two frameworks.
- Workflow fit matters more than brand preference.
- The same quantum idea can often be expressed in both ecosystems, but not with the same level of naturalness.
- Qiskit Machine Learning tends to feel natural when the workflow is centered on kernels, QSVM-like pipelines, or QNN model objects.
- PennyLane tends to feel natural when the workflow is centered on QNodes, explicit measurements, and differentiable hybrid models.
- The purpose of the slide is not to force a winner, but to normalize tool choice as a modeling decision tied to the surrounding workflow.



Learning Tasks: The Main Families

- Supervised learning asks: can we predict labels or values from examples?
- Unsupervised learning asks: can we discover structure without target labels?
- Generative learning asks: can we model a distribution and sample from it?
- Reinforcement learning asks: can an agent improve decisions through interaction and reward?
- This division is useful in the tutorial because each family changes what the output means, what success means, and which metrics make sense.
- The tutorial keeps generative as its own family, even though some broader taxonomies place it under unsupervised learning, because its goals and evaluation logic deserve separate treatment.
- Reinforcement learning is acknowledged as a real ML family, but it is not a main focus of this tutorial.



Why This Tutorial Separates Generative Models

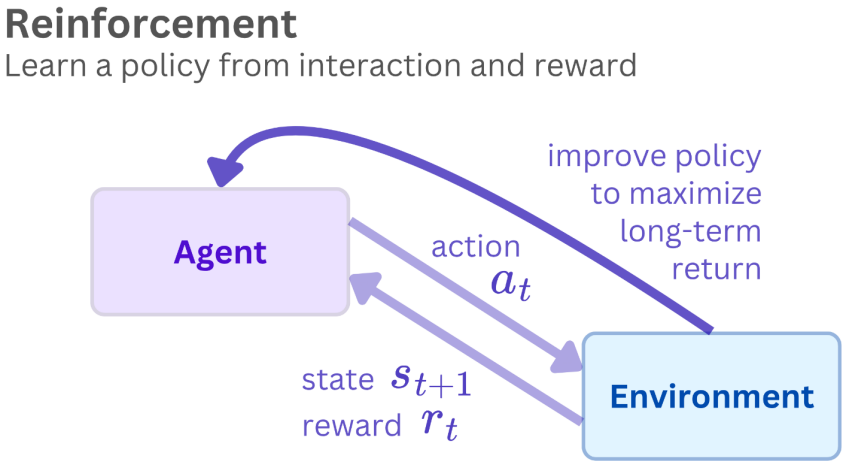
- Unsupervised learning is often about discovering structure in unlabeled data.
- Generative learning is about modeling a distribution and sampling from it.
- Those are related ideas, but not the same objective.
- The tutorial separates generative models because their outputs, questions, and evaluation logic differ enough to deserve their own section.

	Unsupervised	Generative
	discover structure without labels	model a distribution and sample from it
Typical question	What structure or grouping is already in the data?	Can the model reproduce the data distribution?
Typical output	clusters, embeddings, latent structure	samples, likelihood surrogates, distribution scores

$$p_{model}(\bar{x}) \approx p_{data}(\bar{x})$$

23 Reinforcement Learning Is Real ML, but Not Our Focus Here

- Reinforcement learning belongs in the big ML map, but it is not the focus of this section.
- In reinforcement learning, an agent interacts with an environment, takes actions, receives rewards, and improves a policy over time.
- We mention it here only to place it correctly in the ML landscape before narrowing the rest of the tutorial to other families.



24 Metrics Must Match the Task

- There is no single “best metric” across all task families.
- The output type determines what a sensible score looks like.
- A useful rule is:
- if the model predicts labels or scores, choose metrics that reflect the classification goal;
- if it predicts continuous values, choose metrics that reflect error size;
- if it discovers structure without labels, use metrics that reflect cohesion or separation;
- if it models distributions, use metrics that compare distributions rather than single labels.
- Metric choice should follow the question you care about, not habit or convenience.

Choose Metrics From the Learning Objective

In QML, the useful metric depends on what the model is supposed to output: labels, values, clusters, or samples

Classification

Output	class label or confidence score
Metrics	Accuracy, Precision, Recall, F1, ROC-AUC
Typical use	binary or multiclass prediction
Watch out	class imbalance and threshold choice

Regression

Output	real-valued prediction
Metrics	AE, MSE, RMSE, R^2
Typical use	estimating a continuous quantity
Watch out	units, scale, and sensitivity to outliers

Unsupervised Tasks

Output	clusters, embeddings, or latent structure
Metrics	Silhouette, Davies-Bouldin, ARI if labels exist
Typical use	clustering, dimensionality reduction, anomaly search
Watch out	there is rarely a single universal score

Generative Modeling

Output	distribution estimate or samples
Metrics	KL, JS, TV distance, MMD
Typical use	sampling, density modeling, distribution matching
Watch out	metric choice depends on probabilities vs samples

Reinforcement learning is usually evaluated with return, cumulative reward, regret, or success rate, but it is outside the scope of this tutorial

25 Metrics Encode Priorities

- A metric is not neutral because it rewards some behaviors and penalizes some errors more than others.
- In classification, the important distinction may be false positives versus false negatives.
- In regression, the important distinction may be whether large errors should be punished much more strongly than small ones.
- So choosing a metric means choosing what "good performance" should mean for the task.

Precision

useful when false positives are specially costly

Precision emphasizes the quality of predicted positives

Recall

Useful when false negatives are specially costly

Recall emphasizes coverage of actual positives

MAE vs. MSE

both measure prediction error, but they punish mistakes differently

MAE treats errors linearly

MSE punishes large errors more strongly

Metrics are not neutral because they define which tradeoffs the model rewards

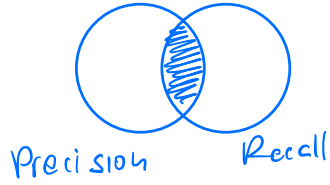
26 Three Metrics Worth Understanding Visually

- Accuracy:
- the fraction of correct predictions,
- useful when classes are reasonably balanced,
- but potentially misleading when one class dominates.
- F1:
- combines precision and recall into one score,
- useful when both false positives and false negatives matter,
- especially when class imbalance makes raw accuracy too optimistic.
- MAE vs MSE:
- both summarize regression error,
- but MSE amplifies large mistakes more strongly than MAE,
- so the choice depends on whether large deviations deserve extra punishment.

Accuracy

	Pred 0	Pred 1
True 0	50	2
True 1	1	43

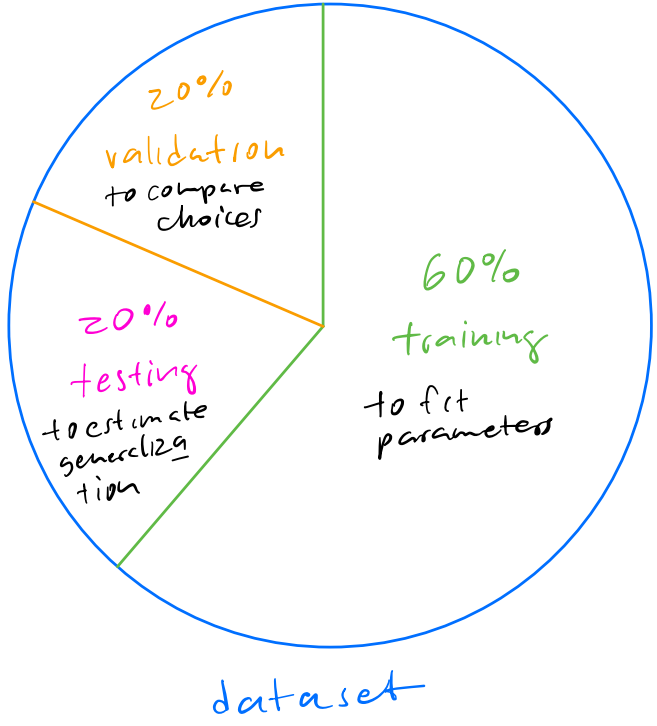
F1



MAE vs MSE

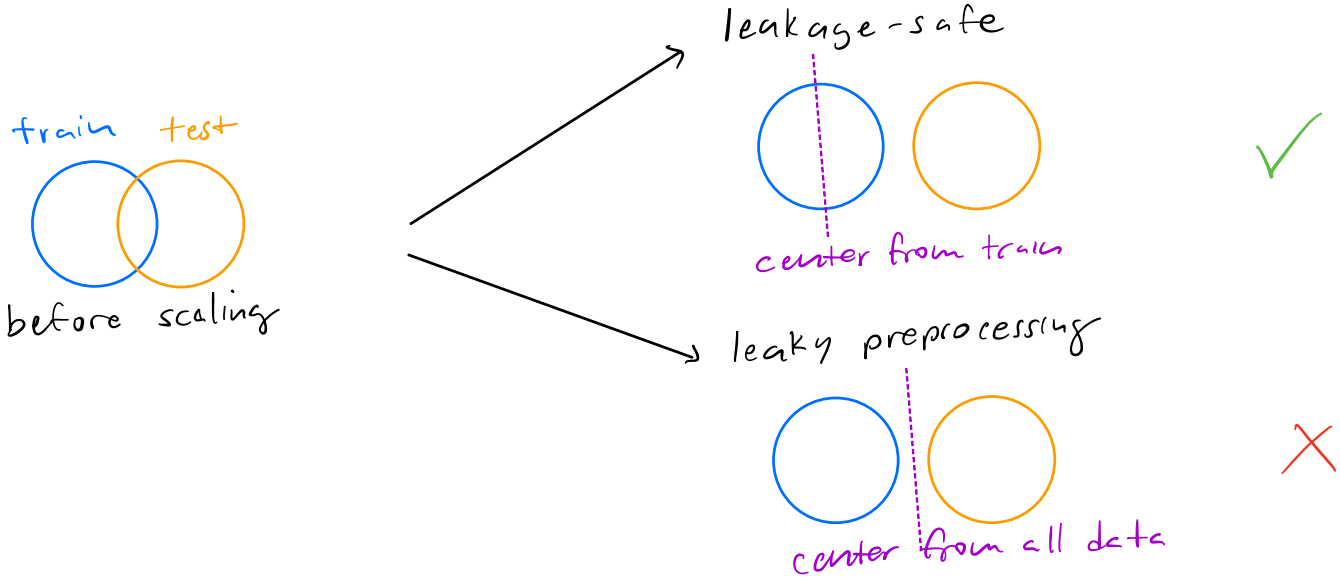
27 Why We Need Train, Validation, and Test

- Train is where model parameters are fitted.
- Validation is where design choices are compared:
- hyperparameters,
- thresholds,
- model families,
- and other decisions that should not touch the test set.
- Test is reserved for the final untouched estimate of generalization.
- The reason for three splits is not bureaucracy:
- without validation, test data gets reused for tuning;
- without test, there is no clean final evaluation.
- Typical proportions such as 60/20/20 or 70/15/15 can be reasonable, but the exact split depends on dataset size and task constraints.



28 Leakage-Safe Thinking

- Split before preprocessing.
- Fit the preprocessor on train only.
- Apply it unchanged to validation and test.
- The reason is not only procedural correctness.
- If the preprocessor sees validation or test data while being fitted, information from those sets leaks into the transformation itself.
- That can make evaluation look cleaner or easier than it really is, even before the model is trained.



Logistic Regression

- Logistic regression is a probabilistic classifier for binary labels.
- It starts from a linear score

$$\vec{w}^\top \vec{z} + b$$

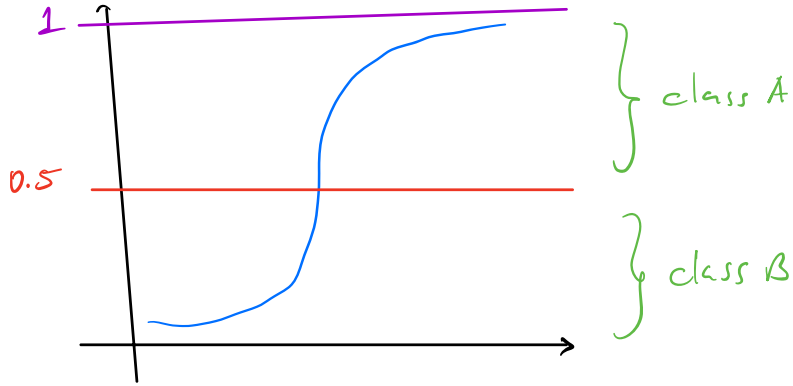
and passes that score through a sigmoid to obtain a value between 0 and 1.

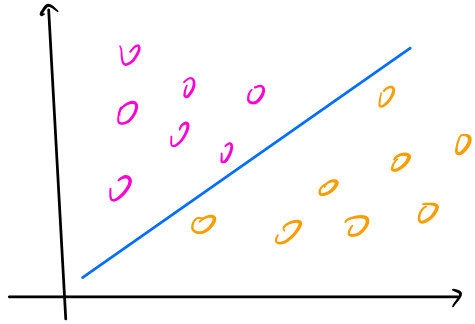
- That value is interpreted as

$$p_\theta(y = 1 | \vec{z})$$

not yet as a hard class label.

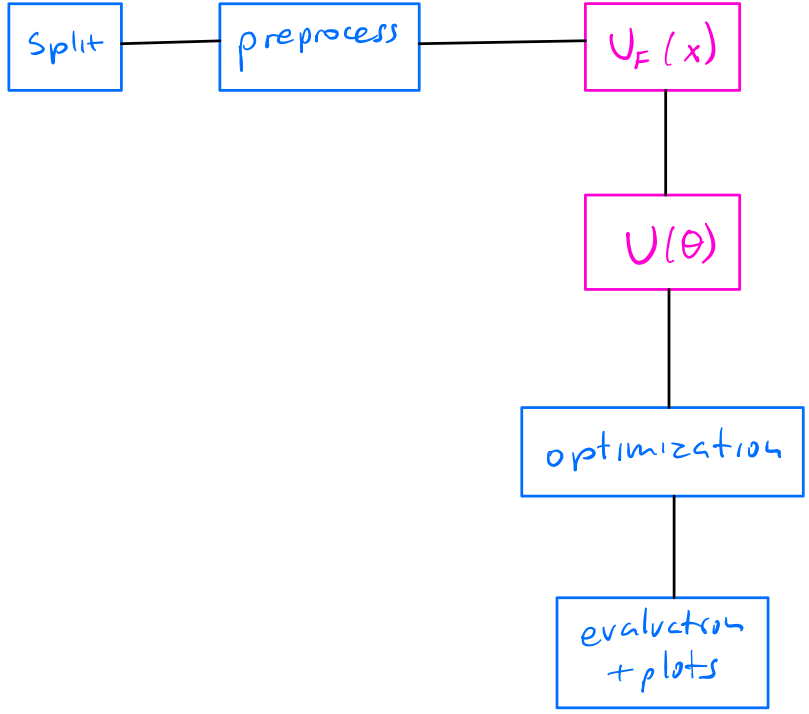
- A decision boundary appears only after choosing a threshold, such as 0.5, to turn probability into $\hat{y}$
- This is why the same model can be understood both as a probability model and as a linear classifier.





30 What the Tiny Variational Preview Adds

- The tiny variational preview is the first place where the tutorial instantiates a genuinely quantum part of the workflow.
- It does not replace the pipeline around it:
  - dataset split,
  - preprocessing,
  - optimization logic,
  - and evaluation remain classical.
- What it adds is a concrete example of the two central quantum blocks:
  - quantum encoding,
  - and quantum model plus measurement.
- The point of the slide is to show where the quantum core sits inside the larger hybrid pipeline.



$$|\psi_\theta(z)\rangle = R_y(\theta)\, R_z(z_2)\, R_y(z_1)\, |0\rangle$$

31 Why the Gate Choices Are Not Yet the Main Point

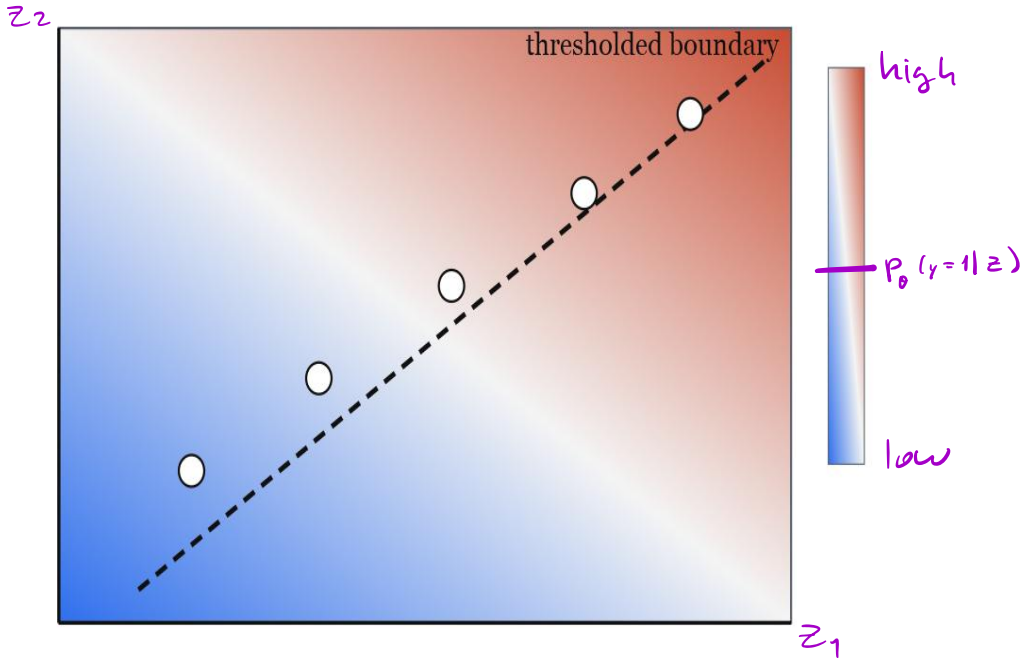
- It is normal at this point not to know yet why certain gates are chosen in a quantum model.
- The tutorial is already showing simple circuit examples, but the detailed reasons for those choices come later.
- The correct early habit is to ask:
- what is the input,
- what is the output,
- what is the metric,
- what is the baseline,
- what modeling hypothesis is being tested.

From Quantum Score to Hard Prediction

- The quantum branch does not directly output a class label.
- It first produces a continuous classical quantity:
- for example an expectation value or a probability.
- The hard prediction appears only after that quantity is converted and thresholded.
- This slide should feel close to the notebook example: first a probability map over the input space, then a decision rule.

$$s_\theta(z) = \langle Z \rangle$$

$$p_\theta(y=1 \mid z) = \frac{1 - s_\theta(z)}{2}$$



$$\hat{y} = \mathbb{1}\left[p_\theta(y=1 \mid z) \geq 1/2 \right]$$

Validation, Test, and the Model Report Card

- A useful model summary is not just one scalar score.
- The point of a report card is to gather the most decision-relevant evidence in one compact view.
- A report card should include:
  - metrics,
  - confusion matrix,
  - short interpretation,
  - limitation,
  - next step.
- This is not a formal standard name; it is a useful presentation habit for reading model results responsibly.

Report card

Metrics

val. accuracy
val. F1
test accuracy
test F1

Confusion matrix

	pred 0	pred 1
true 0		
true 1		

Interpretation

–
–

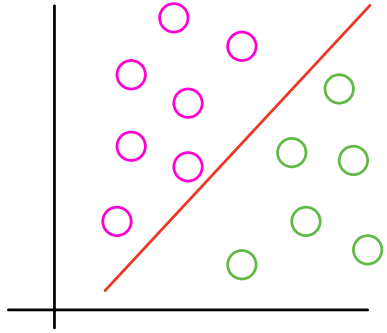
Next step

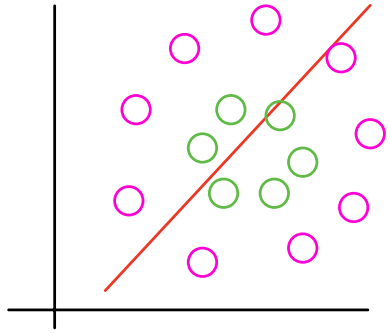
–
–

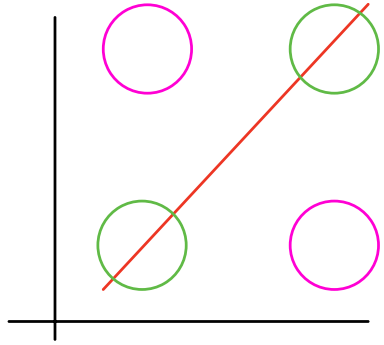
(compact reading tool: score, error pattern, interpretation, next action)

Why Stress-Testing the Baseline Matters

- The goal is not to make the model fail for drama.
- The goal is to separate two different possibilities:
- the pipeline is wrong,
- or the model family is too simple for the geometry of the problem.
- By keeping the protocol fixed and changing the data geometry, we learn whether the limitation comes from methodology or from model expressivity.
- Linear blobs are a case where a linear baseline is structurally compatible.
- Concentric circles and XOR-like blobs are cases where the same baseline is structurally mismatched.
- So the lesson is no longer abstract: some datasets are valuable precisely because they reveal what a model family cannot represent.







35 Why This Makes QML Interesting

- The purpose of this slide is to act as a bridge from foundations to the rest of the course.
- Section 1 does not yet deliver full QML models; it clarifies the questions that must be in place before those models are meaningful.
- Once baseline limits are visible, it becomes meaningful to ask what kind of richer quantum ingredient to introduce next.
- That is why the tutorial now branches into encoding, variational circuits, training and evaluation, kernels, generative models, and QNNs.

Later sections

encoding
variational circuits
training and evaluation

Kernels and QSVM
QNNs
generative models

Closing Summary and Transition to Section 2

- QML extends ML; it does not replace ML fundamentals.
- Framework choice depends on workflow.
- Task family determines sensible metrics.
- Leakage-safe baselines are mandatory.
- Baseline failure motivates richer models, including quantum ones.